

Borrelia burgdorferi, a gram-negative bacterium, is one of the pathogens that cause Lyme disease. *Borrelia* species do not have many genes responsible for synthesis of various amino acids, fatty acids, and nucleotides and therefore are dependent upon infecting a host that can provide those biomaterials for them. To facilitate nutrient uptake, many bacteria express porin proteins, large β -barrel proteins that form wide-diameter, nonspecific pores in the bacterial outer membrane that various molecules can diffuse through. Researchers isolated the outer membrane of *B. burgdorferi* and investigated the proteins associated with it.

Experiment 1

The researchers analyzed the outer membrane proteins by two-dimensional polyacrylamide gel electrophoreses (2D-PAGE). The first dimension was run as blue-native PAGE (BN-PAGE), as seen in Figure 1A. In BN-PAGE the protein samples are pre-treated with Coomassie, an anionic dye that binds to proteins without denaturing them. Afterward, the researchers excised the lane containing the electrophoresed proteins and subjected the whole lane to SDS-PAGE followed by Coomassie staining for the second dimension (Figure 1B).

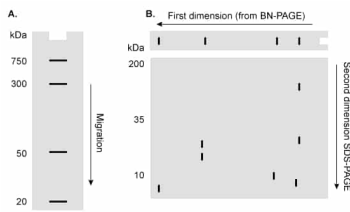


Figure 1 *B. burgdorferi* outer membrane proteins analyzed by 2D-PAGE. Proteins were first analyzed by BN-PAGE for the first dimension (A) and the BN-PAGE lane was rotated and loaded onto an SDS-PAGE gel for the second dimension (B)

Experiment 2

The researchers then purified each complex and recorded each complex's circular dichroism (CD) spectrum. CD spectroscopy measures the difference in absorbance of left- and right-handed circularly polarized light and can be used to analyze the secondary structure of proteins (Figure 2).

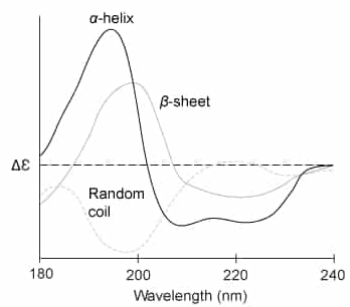


Figure 2 Representative CD spectra of proteins with 100% α -helix, 100% β -sheet, or 100% random coil secondary structures

Experiment 3

Researchers selected the P13 complex for further study. They constructed lipid bilayers containing P13 pores in buffered 1 M KCl and then measured the electrical current and conductance upon applying a voltage across the membrane. Next, they added nonelectrolytes (NEs) of various hydrodynamic radii. Although NE movement through the pores cannot be measured electrically, NEs can compete with the ions for space in the pore, reducing measured pore conductance. Large NEs that cannot enter the pore do not affect conductance through the pore (Table 1).

Table 1 Electrical Conductance of 1M KCl Through P13 Pores in the Presence of Nonelectrolytes

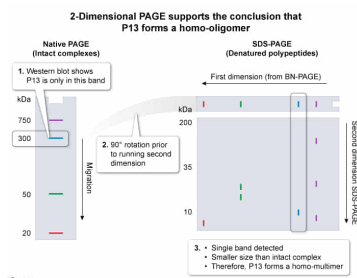
Nonelectrolyte	Molar mass (g/mol)	Hydrodynamic radius (nm)	Conductance ($G\Omega^{-1}$)
None	—	—	0.6
Ethylene glycol	62	0.26	0.3
Glycerol	92	0.31	0.3
Sorbitol	182	0.39	0.2
PEG-200	200	0.50	0.3
PEG-300	300	0.60	0.2
PEG-400	400	0.70	0.6
PEG-600	600	0.80	0.6
PEG-1000	1000	0.94	0.6

If a western blot showed P13 only in the 300 kDa band of the BN-PAGE gel, then the evidence best supports the conclusion that P13:

- ☒ A. forms a homo-multimeric complex. [38%]
- ☐ B. functions as a monomer. [34%]
- ☐ C. is the only outer membrane protein detected. [12%]
- ☐ D. forms a hetero-multimeric complex. [14%]

Correct
38% answered correctly

Explanation:



Protein electrophoresis can be generally classified as either **native or denaturing**. In the passage, both variations were combined to create a two-dimensional gel. The first dimension was a **native page gel** (ie, BN-PAGE), which separates proteins **without denaturing** them. In native gels,

protein complexes generally remain intact because **quaternary structure** is preserved. In the native gel described in the passage, several protein bands were observed, indicating **multiple proteins are present in the outer membrane of *Borrelia burgdorferi*** (**Choice C**); however, P13 was detected in only one band (ie, the 300-kDa band).

SDS-PAGE (ie, **sodium dodecyl sulfate**–polyacrylamide gel electrophoresis), which is a **denaturing** variation of protein electrophoresis, was performed as the second dimension of the 2D electrophoresis experiment. In this experiment, a lane from the BN-PAGE dimension was loaded as the sample for the SDS-PAGE dimension. **Each band** from native PAGE then **formed its own lane** as the constituent proteins were denatured and separated.

Focusing only on the single lane formed by the 300-kDa P13-containing complex, **only one protein band is seen on the SDS-gel**, indicating that **only one peptide subunit** (~13 kDa) contributed to the complex shown in the 300 kDa band of the BN-gel (**Choice D**). The **large size difference** noted before and after denaturation (ie, ~300 kDa vs. ~13 kDa) indicates that several copies of a **single 13-kDa polypeptide join as a homo-multimer to form the 300 kDa complex**.

(Choice B) If P13 functioned as a monomer, then its size after native PAGE would approximately match its size after SDS-PAGE because monomers have no quaternary structure. Instead, the size of P13 after native PAGE is much larger than its SDS-PAGE size, indicating P13 must form a complex.

Educational objective:

Polyacrylamide gel electrophoresis (PAGE) can be classified as native or denaturing. Native PAGE keeps protein structure and complexes intact whereas denaturing page (eg, SDS-PAGE) denatures structure. The output of native PAGE can be run as the input of SDS-PAGE to perform a two-dimensional PAGE experiment.

Time spent: 0 sec
Subject:
Biochemistry

QID: 404009
System:
Amino Acids and Proteins

Borrelia burgdorferi, a gram-negative bacterium, is one of the pathogens that cause Lyme disease. *Borrelia* species do not have many genes responsible for synthesis of various amino acids, fatty acids, and nucleotides and therefore are dependent upon infecting a host that can provide those biomaterials for them. To facilitate nutrient uptake, many bacteria express porin proteins, large β -barrel proteins that form wide-diameter, nonspecific pores in the bacterial outer membrane that various molecules can diffuse through. Researchers isolated the outer membrane of *B. burgdorferi* and investigated the proteins associated with it.

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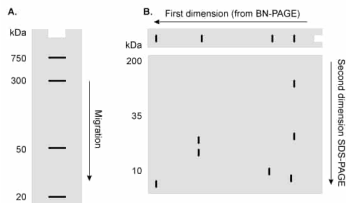


Figure 1 *B. burgdorferi* outer membrane proteins analyzed by 2D-PAGE. Proteins were first analyzed by BN-PAGE for the first dimension (A) and the BN-PAGE lane was rotated and loaded onto an SDS-PAGE gel for the second dimension (B)

Experiment 2

The researchers then purified each complex and recorded each complex's circular dichroism (CD) spectrum. CD spectroscopy measures the difference in absorbance of left- and right-handed circularly polarized light and can be used to analyze the secondary structure of proteins (Figure 2).

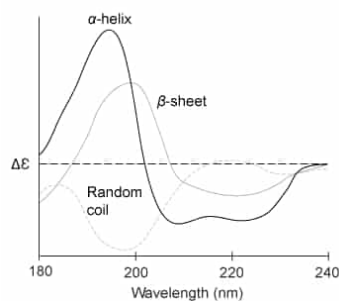


Figure 2 Representative CD spectra of proteins with 100% α -helix, 100% β -sheet, or 100% random coil secondary structures

Experiment 3

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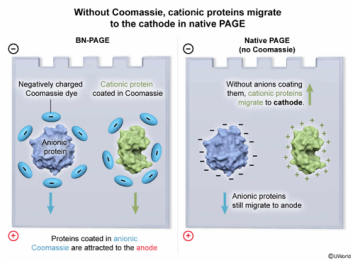
Table 1 Electrical Conductance of 1M KCl Through P13 Pores in the Presence of Nonelectrolytes

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Compared to proteins analyzed by BN-PAGE, proteins analyzed by native PAGE without Coomassie pre-treatment:

- ☐ A. are more likely to be denatured and nonfunctional. [30%]
- ☐ B. are more easily visualized directly after electrophoresis. [13%]
- ☐ C. migrate based only on protein size but not protein shape. [21%]
- ☒ D. may migrate in the opposite direction. [34%]

Explanation:



Electrophoresis is a common experiment in biochemistry that **separates molecules** by **size, shape, and/or charge**. An external electric field induces molecules to migrate through a porous gel matrix. **Negatively** charged molecules migrate to the **positive electrode** (ie, the **anode** of an electrolytic cell), and smaller molecules (eg, lower molecular weight, more compact globular shape) are less impeded by the gel matrix and migrate more quickly than larger molecules.

In **sodium dodecyl sulfate**–polyacrylamide gel electrophoresis (SDS-PAGE), the **anionic** detergent sodium dodecyl sulfate serves two purposes: it **coats proteins with a negative charge** allowing them to migrate toward the anode, and it also **denatures** proteins. Removing the SDS allows proteins to retain native conformation, but it also allows proteins to retain their native charge. Therefore, proteins whose **isoelectric point** is above the pH of the running buffer will be **positively charged** and **migrate in the opposite direction**, toward the negatively charged cathode.

In the passage, a variation of native page that includes Coomassie dye, BN-PAGE, is used. The passage states that Coomassie is an **anionic**, but not denaturing, dye that binds to proteins. The inclusion of this **anionic dye allows proteins to migrate toward the anode**, regardless of the protein's isoelectric point. Therefore, cationic proteins that are not treated with Coomassie will migrate the opposite direction from those that are.

(Choice A) Native PAGE gels by definition are designed to retain the native conformation of proteins. Therefore, proteins run under native conditions are unlikely to be denatured regardless of the inclusion of Coomassie.

(Choice B) Proteins generally *need to be stained* for visualization after electrophoresis. Pre-treatment *with Coomassie dye* would make proteins easier to visualize than proteins not treated with Coomassie.

(Choice C) Under native PAGE conditions, *both size and shape* contribute to the migration of proteins through the gel matrix, regardless of the inclusion of Coomassie.

Educational objective:

Electrophoresis separates molecules by size, shape, and charge. In SDS-PAGE, sodium dodecyl sulfate coats proteins with a negative charge allowing all proteins to migrate to the anode. In native PAGE, anionic dyes like Coomassie serve a similar purpose without denaturing proteins.

Time spent:0 sec
Subject:
Biochemistry

QID:404010
System:
Amino Acids and Proteins

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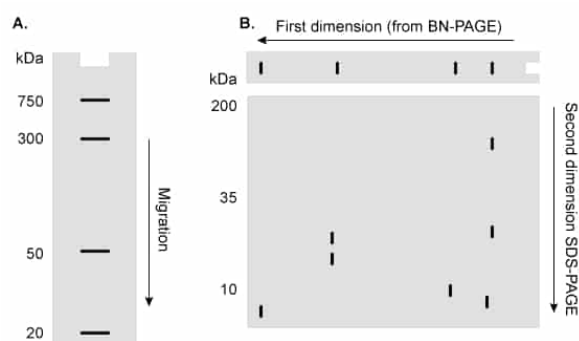


Figure 1 *B. burgdorferi* outer membrane proteins analyzed by 2D-PAGE. Proteins were first analyzed by BN-PAGE for the first dimension (A) and the BN-PAGE lane was rotated and loaded onto an SDS-PAGE gel for the second dimension (B)

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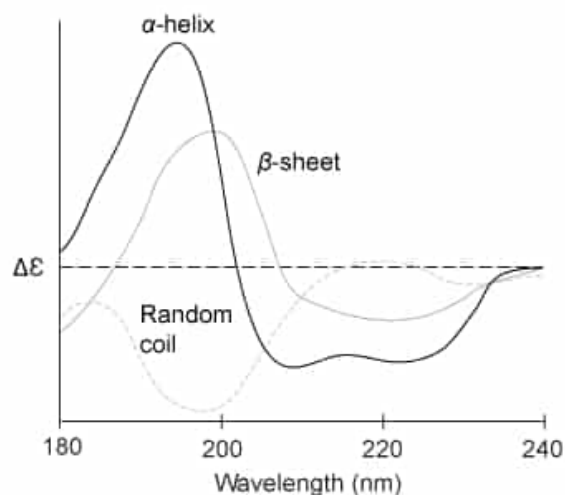


Figure 2 Representative CD spectra of proteins with 100% α -helix, 100% β -sheet, or 100% random coil secondary structures

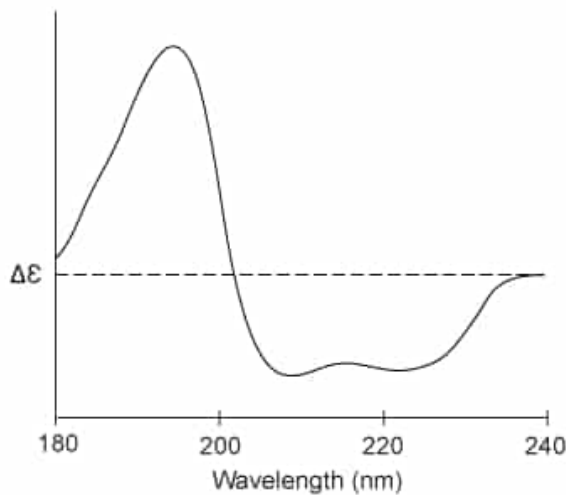
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The following CD spectrum was obtained for P13:



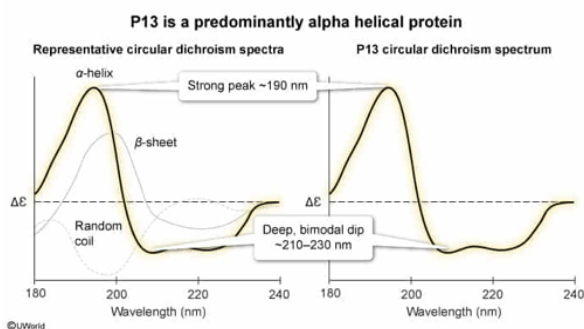
Based on this, the predominant secondary structure element in P13 is:

- ✓ ☒ A. α -helix. [93%]
☐ B. β -sheet. [3%]
☐ C. random coil. [1%]
☐ D. a roughly equal proportion of β -sheet and random coil. [1%]

Correct

94% answered correctly

Explanation:



Circular dichroism (CD) is a spectroscopic technique that uses **circularly polarized light** to study the **secondary structure** of proteins. Different secondary structure elements absorb right-handed and left-handed circularly polarized light of a given wavelength differently. This difference is recorded as the difference in ellipticity ($\Delta\epsilon$). Figure 2 of the passage shows the representative CD spectra of proteins with 100% α -helical secondary structure, 100% β -sheet secondary structure, and 100% random coil secondary structure.

The question shows the CD spectrum of the P13 protein and asks which secondary structural element is predominantly shown. This can be discerned by comparing the P13 spectrum to the spectra shown in Figure 2. The P13 spectrum displays a **strong peak at ~190 nm** and a **deep, bimodal dip from ~210 to 230 nm**. Of the

representative spectra shown in Figure 2, this pattern **best matches the 100% α -helix spectrum**. Therefore, P13 consists predominantly of α -helices.

(Choice B) Although the passage states that *most* porins are β -barrel proteins, the P13 spectrum does *not* match the β -sheet secondary structure. A β -sheet structure would be expected to have a peak at ~195 nm and a shallow unimodal dip centered at 220 nm.

(Choice C) A predominant structure of random coils would have a negative peak at ~195 nm.

(Choice D) An equal mixture of β -sheet and random coil would be expected to look like the sum of the 100% β -sheet and 100% random coil spectra. The positive and negative peaks at ~195 nm would cancel each other, leaving a relatively flat spectrum with a very shallow dip at 220 nm.

Educational objective:

Circular dichroism (CD) spectroscopy is a technique that can be used to study the secondary structure of proteins. The CD spectrum of a given protein can be compared to standard spectra to elucidate the secondary structure features of the sample protein.

Time spent:0 sec

Subject:
Biochemistry

QID:404011

System:
Amino Acids and Proteins

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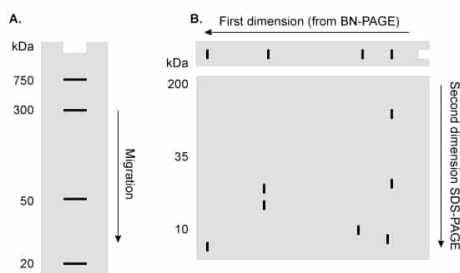


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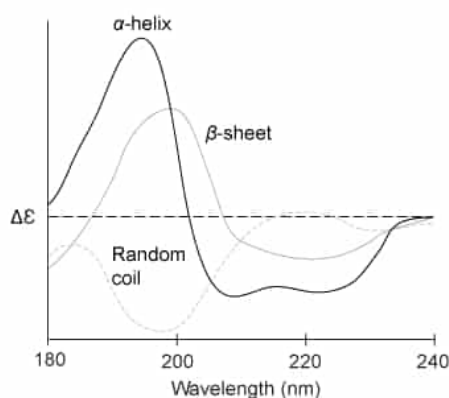


Figure 2 Representative CD spectra of proteins with 100% α -helix, 100% β -sheet, or 100% random coil secondary structures

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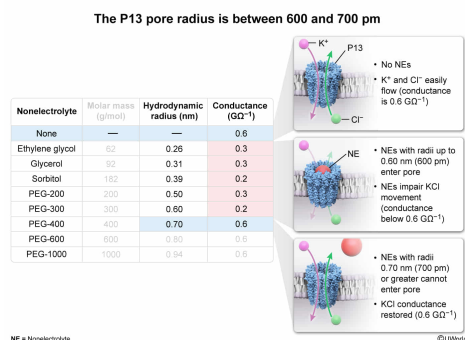
Based on the information given in the passage, the radius of the P13 channel pore can be best described as being:

- ☐ A. between 300 and 400 pm [24%]
☐ B. between 400 and 500 pm [20%]
☒ C. between 600 and 700 pm [42%]
☐ D. greater than 1,000 pm [12%]

Correct

43% answered correctly

Explanation:



Porins are large, nonspecific pores in a lipid bilayer that allow for the passage of small molecules and ions across the membrane. If an electric potential is induced across the membrane, charged ions will travel down their electric gradients, and the movement of these charged ions results in an electric current that can be measured. The passage describes an experiment to measure the pore diameter of these porins by measuring the effect of nonelectrolytes (NEs) on **how easily these charged ions can pass** (ie, by measuring **conductance**).

In the **absence of NEs**, conductance is at a **baseline level**. Table 1 shows that this level is $0.6 \text{ G}\Omega^{-1}$. As NEs are included, the NEs enter the pore, impeding movement of the charged ions. Therefore, **conductance decreases**. Table 1 shows that this happens with NEs as large as 0.60 nm (ie, **600 pm**) in radius (ie, PEG-300). Therefore, the pore radius must be **bigger than 600 pm**.

PEG-400, however, which has a radius of 0.70 nm (700 pm), **does not lower conductance from baseline**. The conductance with PEG-400 is $0.6 \text{ G}\Omega^{-1}$. The passage states that this occurs if the NE is **too big to enter the pore** and therefore does not impede ion movement. Because a 700-pm NE does not lower conductance, the pore radius **is smaller than 700 pm**.

Taken together, the experiments show that the **pore radius is between 600 and 700 pm**.

(Choice A) 300 and 400 are the *molecular weights* of the largest NE that can enter and the smallest NE that cannot enter the pore, respectively. However, the question asks for the *radius*, not the molecular weight.

(Choice B) PEG-200, which has a radius of 500 pm, is still small enough to enter the pore. Therefore, the radius of the pore must be above 500 pm.

(Choice D) NEs as small as 700 pm in radius no longer impede channel conductance, which the passage states is due to the inability of these large NEs to enter the pore. Therefore, the pore must be smaller than 700 pm in radius.

Educational objective:

Porins form wide-diameter, nonspecific pores in a membrane. By analyzing data showing the ability of molecules to impede the flow of ions through a porin, researchers can estimate the diameter and radius of a porin.

Time spent:0 sec

Subject:
Biochemistry

QID:404012

System:
Amino Acids and Proteins